UNIVERSITÉ PARIS 1
PANTHÉON SORBONNE

Quantitative Methods in Finance

Counterfactual Inflation Analysis:
What If Ukraine Had Been Part of the Euro Area?

May 2026

Master 2 – Finance, Technology, Data

Academic Year 2025–2026

Part A — Ukraine's Monetary Regime: A Preliminary Analysis

Deliverable 1 — Chronological Summary Table

Colour legend:

Light blue = dollar peg — Yellow = devaluation/crisis — Green = inflation targeting (genuine sovereignty) — Orange = wartime peg — Grey = wartime flexible IT

Table 1: Chronology of Ukraine's Exchange Rate Regimes

Period	UAH/USD (approx.)	IMF de facto classification (AREAER label)	Key events	Genuine monetary sovereignty?
Jan. 2000 – 29 Apr. 2008	5.44 (2000) → 5.05 (2006–2007)	*Conventional pegged arrangement*	NBU maintained the exchange rate within an extremely narrow band through massive FX interventions. De jure "managed floating" since Feb. 2000 with no operational change. IMF (Selected Issues, 2016) explicitly describes this period as a case of *fear of floating* in the sense of Calvo and Reinhart (2002). The EFF Request (No. 15/69) confirms in retrospect that IMF technical assistance for an IT transition was first provided in 2004 but *"until 2014 the exchange rate remained effectively pegged to the dollar while macro-financial imbalances worsened."*	**NO**: monetary policy subordinated to maintaining the dollar peg

Period	UAH/USD	IMF Label	Key Events	Sovereignty?
30 Apr. – 28 Dec. 2008	5.05 → 7.70 (+52.5% dep.)	*Other managed arrangement*	Brief move toward flexibility → capital flight → Emergency FX controls imposed 28 Oct. 2008 (restrictions on FX purchases, limits on early withdrawal of bank deposits). Official fixing at 7.70 on Dec. 29. Approximately −34.4% loss of UAH value. Capital controls on some current-account transactions lifted by May 2010.	**NO**: crisis-driven devaluation; immediate re-fixing
Jan. 2009 – Feb. 2010	∼7.70 – 7.90	*Other managed arrangement*	Post-devaluation transition. NBU stabilised the rate without a formal band. Residual current-account controls lifted by May 2010.	**NO**: de facto re-fixing in progress
Mar. 2010 – Apr. 2014	∼7.99 – 8.15	*Stabilized arrangement*	Return to the ∼8 UAH/USD peg. AREAER 2013 notes stability within 2% band. IMF later confirmed flexibility was "forced and short lived".	**NO**: dollar peg reconstituted
Apr. 2014 – Dec. 2015	8 → ∼25 (−68% value loss)	*Floating (under capital controls)*	Maidan & Crimea. Peg formally abandoned Feb 7, 2014. Hryvnia depreciation of ∼90% by end-2014 relative to 2013. **Capital controls:** 1) 75% mandatory export surrender requirement (Sept. 2014); 2) cap on individual FX purchases; 3) ban on dividend repatriation from OTC securities; 4) policy rate raised to 19.5% then 30% (Feb. 2015). Actual inflation 48% vs. EFF projection of 27%.	**NO**: forced float; reserve depletion; maximum counterfactual treatment intensity

Period	UAH/USD	IMF Label	Key Events	Sovereignty?
Jan. 2016 – 23 Feb. 2022	∼25 – 29	*Floating*	IT formally adopted: monetary policy guidelines approved by NBU Council 21 Dec. 2016. Inflation targets: 12% (2016), 8% (2017), 6% (2018), 5% ±1pp (2019+). Policy rate became the primary instrument; capital controls gradually lifted from 2016. Behaviour did partially change: the policy rate was actively used to counter inflation and the exchange rate was no longer defended. However, credibility remained fragile, expectations persistently above targets (IMF 2016) and wiiw (2017) argues stabilisation in 2016–2018 owed more to capital control liberalisation than to IT itself.	**YES**: only period of genuine discretionary monetary sovereignty
24 Feb. – 20 Jul. 2022	Fixed at 29.2549	*Stabilized arrangement*	Russian invasion. NBU fixed rate via Resolution No. 18. IT suspended. Emergency capital controls.	**NO**: IT suspended; wartime peg
21 Jul. 2022 – 2 Oct. 2023	Fixed at 36.5686	*Stabilized arrangement*	Rate re-fixed via Resolution No. 154. ±2% band maintained. Strategy for return to IT approved June 2023.	**NO**: wartime peg; progressive exit strategy
3 Oct. 2023 – end 2025	36.57 → ∼42.4	*Floating (“managed flexibility”)*	Move to “managed flexibility”. “Flexible IT” framework approved Sept 2024. Point target of 5%. Gradual depreciation.	**LIMITED**: constrained by war and active IMF EFF conditions.

Notes:

[1] Convention on 2008 depreciation: 5.05 → 7.70 is +52.5% (direct quote) or −34.4% (UAH value).

[2] Retroactive reclassification of Feb 2009 reflects methodological modification, not substantive alteration.

Deliverable 2 — Monetary Sovereignty and Implications for the Counterfactual

Ukraine had genuine monetary sovereignty only once: from January 2016 to February 2022, when the NBU operated a real inflation-targeting regime with an explicit target path (12%→8%→6%→5%±1pp) and the policy rate as its primary instrument. Before 2016, the NBU maintained a *de facto* dollar peg despite declaring a managed float, a classic case of fear of floating (Calvo & Reinhart, 2002). The two major devaluations (2008–2009, 2014–2015) were not policy choices but forced adjustments driven by reserve depletion, accompanied by capital controls in both episodes. This matters for the counterfactual because Euro Area membership would have been a near-neutral anchor substitution during peg periods, a very large inflationary shock absorber during the devaluation crises and a genuine replacement of NBU discretion only during 2016–2022.

Part B — The Counterfactual

Identification Strategy

The counterfactual is constructed using two complementary methods, consistent with the methodological requirements of the exam and the regime chronology established in Part A.

Primary method — SVAR Bayoumi-Eichengreen (1993) / Blanchard-Quah (1989). We estimate bivariate structural VARs $[\Delta \log IPI_t, \Delta\pi_t^{YoY}]$ for Ukraine and the Euro Area separately, using IPI from Eurostat (STS_INPR_M, EA20, B-D, seasonally adjusted, 2015=100) as the output variable. Both systems are estimated on 2006–2021 to avoid the structural break introduced by the war. The Blanchard-Quah long-run restriction identifies supply vs. demand shocks: demand shocks have zero long-run effect on output, while supply shocks may have permanent output effects. This operationalises "being in the Euro Area" precisely: Ukraine's structural demand shocks, representing domestic monetary conditions, are replaced by EA demand shocks representing ECB policy, while Ukraine's supply shocks (energy, agriculture, geopolitics) remain unchanged.

The demand shock substitution is implemented via IRF convolution ($H = 24$ months), which correctly accounts for the dissipation of shock effects over time:

$$\delta_t = \sum_{j=0}^{23} \mathrm{IRF}_{\pi,d}(j) \cdot \left(\varepsilon_{t-j}^{d,EA} - \varepsilon_{t-j}^{d,UKR}\right)$$

Both VARs are estimated on standardised data; IRFs are rescaled by $\hat{\sigma}(\Delta\pi^{UKR})$ to recover percent-point inflation units. The SVAR gap isolates the demand-policy component that ECB discipline would have directly replaced: the difference between the CM and SVAR gaps in each sub-period captures the share of the inflationary divergence attributable to the exchange rate and supply channels.

Complement — Ciccarelli-Mojon (2010) factor model. To assess the *amplitude* of the total treatment effect, including the exchange-rate and credibility channels that the SVAR allocates to supply shocks by construction, we complement the SVAR with a dynamic factor model. We extract the Euro Area common inflation factor $\hat{F}_t^{EA}$ from a panel of 11 EA countries' HICP series (Austria, Belgium, Germany, Spain, Finland, France, Greece, Ireland, Italy, Netherlands, Portugal) using Principal Component Analysis fitted on 2006–2019 only (to avoid the 2022 energy shock dominating the first component), then projected onto the full 2006–2025 sample. The factor explains 73.2% of EA inflation variance over the estimation window.

Ukraine's inflation is modelled as:

$$\pi_t^{UKR} = \lambda \cdot F_t^{EA} + \varepsilon_t^{UKR}$$

The loading λ is estimated over the 2018–2021 window (stable IT period), yielding $\hat{\lambda} = 0.683$ ($p = 0.000$, $R^2 = 0.342$), the only sub-period where Ukraine exercised genuine monetary discretion and co-movement with ECB policy is economically meaningful. The robustness table confirms λ is non-significant over earlier dollar-peg windows. The counterfactual is:

$$\hat{\pi}_t^{CF} = \hat{\lambda} \cdot \hat{F}_t^{EA} + \bar{\varepsilon}$$

where $\bar{\varepsilon} = \overline{(\pi^{UKR} - \hat{\lambda}F^{EA})}_{2018\text{–}2021}$ is the structural mean inflation gap. The uncertainty band reflects $\hat{\lambda} = 0.257$ (2016–2021 lower bound). This counterfactual is explicitly not the cross-sectional mean of EA countries' inflation, as required by the exam.

Data Sources

- **Ukraine CPI:** State Statistics Service of Ukraine (SSSU) via SDMX, course repository. Monthly MoM index transformed to YoY: $\pi_t^{YoY} = (P_t/P_{t-12} - 1) \times 100$.
- **EA HICP:** ECB Data Portal, course repository. 11 countries, Jan. 2000–Dec. 2025.
- **UAH/USD exchange rate:** National Bank of Ukraine API, programmatic download.
- **Ukraine IPI:** IMF Production Indexes dataset, 2002–Jun. 2025.
- **Euro Area IPI:** Eurostat SDMX 2.1 REST API, dataset `STS_INPR_M`, geo=EA20, nace_r2=B-D, s_adj=SCA, unit=I15. Downloaded programmatically within the notebook.
- **Commodity prices:** Henry Hub gas (`MHHNGSP`) and wheat (`PWHEAMTUSDM`) from FRED, programmatic download.

Estimation Choices and Diagnostics

- **Stationarity:** ADF tests confirm $\Delta \log IPI_{UKR}$ ($p = 0.0004$) and $\Delta \log IPI_{EA}$ ($p < 0.0001$) are stationary. Both $\Delta\pi$ series are stationary by construction.
- **Lag selection:** BIC selects $p = 6$ for Ukraine (capped from BIC=12), $p = 2$ for EA.
- **VAR stability:** All companion matrix eigenvalues < 1 for both systems (Ukraine: max $|e| = 0.824$; EA: max $|e| = 0.554$).
- **BQ identification:** Supply/demand shock correlation = 0.0000 for both systems, confirming exact orthogonality.
- **Granger causality:** F_t^{EA} does not Granger-cause Ukraine CPI at any lag (all $p > 0.13$), consistent with the structural decoupling finding and justifying the 2018–2021 window for $\hat{\lambda}$.

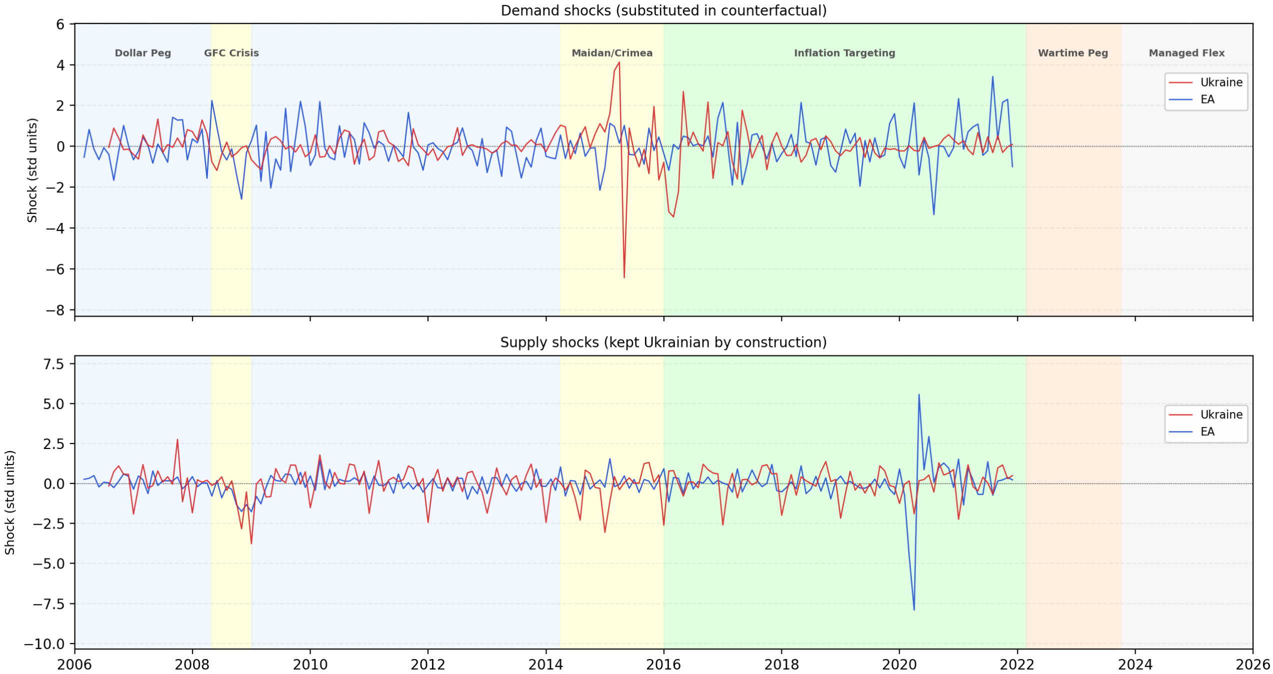

Figure 1: Structural demand and supply shocks identified via Blanchard-Quah decomposition. Top panel: demand shocks substituted in the counterfactual. Bottom panel: supply shocks kept Ukrainian. Crisis periods shaded. The large negative Ukrainian demand shock in 2014–2015 confirms that the devaluation episode was not primarily demand-driven, consistent with the SVAR gap of +3.4pp vs. the CM gap of +24.5pp.

Deliverable 1 — Counterfactual Figure

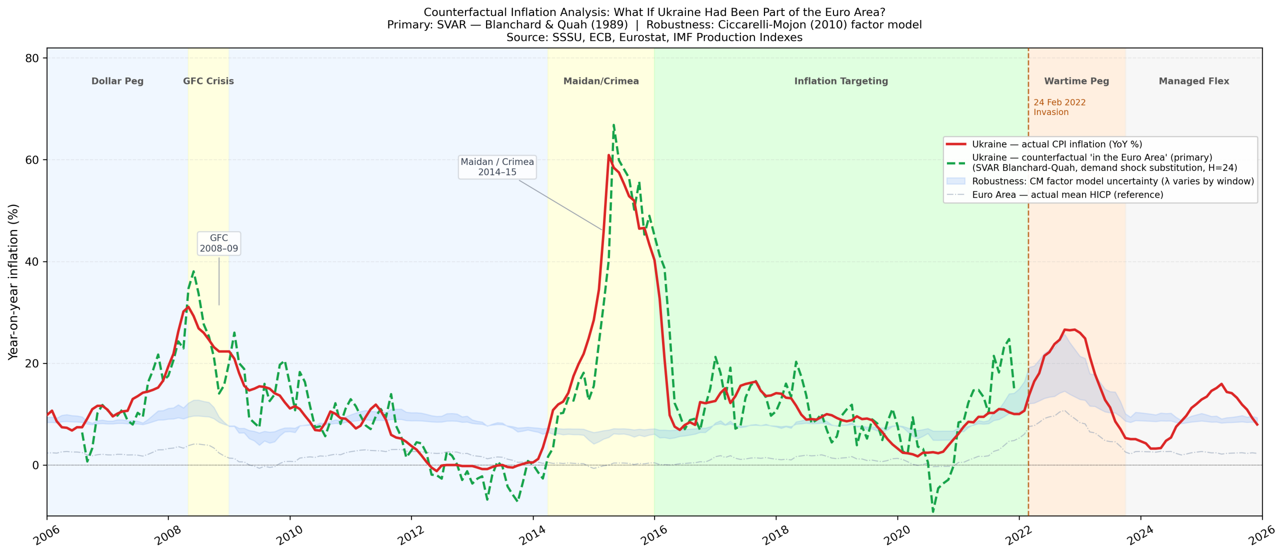

Figure 2: Counterfactual inflation analysis: what if Ukraine had been part of the Euro Area? Background colours correspond to monetary regimes as per the Part A chronology table. *Primary:* SVAR Blanchard-Quah (1989), demand shock substitution, IRF convolution $H = 24$ months. *Complement:* Ciccarelli-Mojon (2010) factor model ($\hat{\lambda} = 0.683$, 2018–2021 window); shaded band reflects $\hat{\lambda} = 0.257$ (2016–2021 lower bound). Sources: SSSU, ECB, Eurostat, IMF Production Indexes.

Sanity check — average gap (actual – counterfactual) by sub-period:

Period	CM gap (pp)	SVAR gap (pp)	Assessment
2006–2013 (peg era)	+1.2	+0.3	Quasi-null: consistent with anchor substitution
2008–09 (GFC)	+12.0	+2.0	Moderate: devaluation pass-through absent under ECB
2014–2015 (deval.)	+24.5	+3.4	Maximum treatment episode: see discussion below
2016–2021 (IT era)	+2.6	−1.5	NBU more restrictive than ECB: see discussion below

On the contrast between CM and SVAR gaps. The divergence between the two methods is itself an analytical result, not a limitation. The SVAR gap (+3.4pp in 2014–2015) captures only the component attributable to *demand shock differences*: by isolating Blanchard-Quah structural demand shocks, it asks a narrower question, was Ukraine's domestic demand intrinsically more inflationary than the EA's? The answer is moderately yes. This implies that the dominant driver of the 2014–2015 inflation spike was *not* excess domestic demand but the exchange rate channel and supply-side pass-through, which the SVAR correctly allocates to the supply shock component kept Ukrainian by construction. The CM gap (+24.5pp) captures the *total* inflationary differential including these channels. The two methods are therefore complementary: the SVAR gap measures the demand-policy component that ECB discipline would have directly replaced; the remainder of the CM gap represents adjustment costs that would have been displaced into internal devaluation rather than eliminated.

Deliverable 2 — Interpretation

Our SVAR counterfactual reveals that monetary sovereignty for Ukraine has functioned as a double-edged sword, where the "benefit" of the exchange rate as a shock absorber came at a massive inflationary cost. During the 2008–2009 and 2014–2015 crises, sovereignty was effectively an illusion because the NBU didn't exercise discretion but suffered forced devaluations that translated into immediate inflationary spikes (+2.0pp and +3.4pp of demand-driven gap but over +20pp of total price divergence). In these episodes, Euro Area membership would have provided the benefit of price stability and imported credibility but at the cost of preventing nominal adjustment, forcing a painful internal devaluation (wage and price compression) similar to the Baltic experience. Conversely, the 2022 crisis highlights the ultimate strategic value of sovereignty, which is why, while it fuelled inflation through monetary financing, it allowed the NBU to act as a lender of last resort for the state's survival, a degree of freedom that Euro Area membership would have strictly prohibited, potentially transforming a liquidity crisis into a terminal sovereign default. Ultimately, Ukraine's sovereignty has been a tool for survival in extreme shocks, but its chronic lack of credibility makes Euro Area membership a net benefit for long-term macroeconomic anchoring.

References

Primary IMF Sources

- **IMF, *Annual Report on Exchange Arrangements and Exchange Restrictions* (AREAER)**, Ukraine country chapters, positions 2000–2024. *elibrary-areaer.imf.org*
- **IMF Country Report No. 10/262**, *Ukraine: Request for Stand-By Arrangement*, July 2010.
- **IMF Country Report No. 11/52**, *Ukraine: First Review Under the Stand-By Arrangement*, December 2010.
- **IMF Country Report No. 14/145 & 14/263** (2014).
- **IMF Country Report No. 15/69**, *Ukraine: Request for Extended Arrangement Under the Extended Fund Facility and Cancellation of Stand-By Arrangement*, March 2015.
- **IMF Country Report No. 16/320**, *Ukraine: Ex Post Evaluation of Exceptional Access Under the 2014 Stand-By Arrangement*, September 2016.
- **Madrid, P. & de Oliveira Lima, I.L.**, "Transitioning to Inflation Targeting and Enhancing Monetary Policy Effectiveness," IMF Selected Issues Paper, 2017 Article IV Ukraine.

Primary NBU Sources

- **NBU Board Resolution No. 18** (24 Feb. 2022) — fixing at 29.2549 UAH/USD.
- **NBU Board Resolution No. 154** (21 Jul. 2022) — re-fixing at 36.5686 UAH/USD.
- **NBU**, *Strategy for Easing FX Restrictions, Transition to Greater Exchange Rate Flexibility and Return to Inflation Targeting*, 29 Jun. 2023. *bank.gov.ua*
- **NBU**, *Monetary Policy Guidelines for the Medium Term*, 11 Sept. 2024. *bank.gov.ua*
- **NBU**, *FX Restrictions and Exchange Rate Policy* (updated 2026). *bank.gov.ua/en/markets/liberalization*

Data Sources

- **State Statistics Service of Ukraine (SSSU)**, Ukraine CPI monthly series via SDMX, course repository.
- **ECB Data Portal**, Euro Area HICP panel, 11 countries, 2000–2025, course repository.
- **Eurostat**, Industrial Production Index (STS_INPR_M), EA20, nace_r2=B-D, SCA, 2015=100. Downloaded programmatically via SDMX 2.1 REST API.
- **IMF Production Indexes dataset**, Ukraine IPI, 2002–2025-M06.
- **National Bank of Ukraine API**, UAH/USD daily exchange rate, 2000–2025.
- **FRED (Federal Reserve Bank of St. Louis)**, Henry Hub gas price (`MHHNGSP`), wheat price (`PWHEAMTUSDM`).

Academic and Institutional Sources

- **Astrov, V. & Podkaminer, L. (2017).** *Ukraine: Selected Economic Issues.* Vienna Institute for International Economic Studies (wiiw), Policy Notes and Reports No. 19, December 2017.
- **Barro, R.J. & Gordon, D.B. (1983).** Rules, Discretion and Reputation in a Model of Monetary Policy. *Journal of Monetary Economics*, 12(1):101–121.
- **Bayoumi, T. & Eichengreen, B. (1993).** Shocking aspects of European monetary integration. In Torres, F. & Giavazzi, F. (eds.), *Adjustment and Growth in the European Monetary Union*, pp. 193–229. Cambridge University Press.
- **Blanchard, O.J. & Quah, D. (1989).** The dynamic effects of aggregate demand and supply disturbances. *American Economic Review*, 79(4):655–673.
- **Calvo, G.A. & Reinhart, C.M. (2002).** Fear of Floating. *Quarterly Journal of Economics*, 117(2):379–408.
- **Ciccarelli, M. & Mojon, B. (2010).** Global inflation. *Review of Economics and Statistics*, 92(3):524–535.
- **De Grauwe, P. (2012).** The governance of a fragile eurozone. *Australian Economic Review*, 45(3):255–268.
- **Frankel, J.A. & Rose, A.K. (1998).** The endogeneity of the optimum currency area criteria. *Economic Journal*, 108(449):1009–1025.
- **Giavazzi, F. & Pagano, M. (1988).** The advantage of tying one's hands: EMS discipline and central bank credibility. *European Economic Review*, 32(5):1055–1075.
- **Kilian, L. & Lütkepohl, H. (2017).** *Structural Vector Autoregressive Analysis.* Cambridge University Press.
- **Mundell, R.A. (1961).** A Theory of Optimum Currency Areas. *American Economic Review*, 51(4):657–665.